A finite counterexample to ideal-cardinality homomesy on constant-parameter fences

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Abstract

We give an explicit finite counterexample to Conjecture 5.4(b) of Elizalde, Plante, Roby, and Sagan, *Rowmotion on fences*. For the 32-element fence $\breve{F}(3^{11})$, the lower ideal $\{5, 6, 7, 8, 12\}$ has rowmotion period 237, and the sum of ideal cardinalities on its orbit is 3774. The orbit average is therefore 1258/79, rather than 16. We supply the complete orbit, an independent integer-arithmetic checker, and a Lean 4 formalization connecting the paper's segment convention to the orbit calculation. The formalization uses `native_decide`; its trust assumptions include `Lean.ofReduceBool` and the correctness of the Lean compiler. No claim of priority is made.

1 The conjecture and the precise fence

For a finite poset P, write $\mathcal{J}(P)$ for its set of lower ideals. Rowmotion on lower ideals is

$$\rho(I) = {\downarrow}\min(P \setminus I).$$

A statistic f is c-mesic if its average on every rowmotion orbit is c. Conjecture 5.4(b) of [1, p. 35] states that, for a fence $\breve{F}(a^t)$ with constant segment parameter and an odd number t of segments, the statistic $I \mapsto |I|$ is $|P|/2$-mesic. Segments here are maximal chains. The parameter of a segment is one plus the number of its *unshared* vertices, not uniformly its number of edges; a shared vertex belongs to two adjacent segments [1, p. 17].

Let P have vertices $1, \ldots, 32$. Its cover relations are given by the following chains, each written from smaller to larger in the poset:

$$\begin{array}{lll} 1 < 2 < 3, & 6 < 5 < 4 < 3, & 6 < 7 < 8 < 9, \\ 12 < 11 < 10 < 9, & 12 < 13 < 14 < 15, & 18 < 17 < 16 < 15, \\ 18 < 19 < 20 < 21, & 24 < 23 < 22 < 21, & 24 < 25 < 26 < 27, \\ 30 < 29 < 28 < 27, & 30 < 31 < 32. & \end{array}$$

Equivalently, the underlying cover graph is the path $1, \ldots, 32$. For $1 \le e \le 31$, its edge from e to $e+1$ points upward exactly when $\lfloor e/3 \rfloor$ is even. The partial order is the reflexive transitive closure of these oriented edges.

The displayed chains are exactly the maximal chains: a chain can follow the oriented path only until the next change of direction. Their shared vertices are $3, 6, 9, \ldots, 30$. Each endpoint segment has three vertices and one shared vertex, whereas each of the nine internal segments has four vertices and two shared vertices. Thus every segment has exactly two unshared vertices and parameter 3. There are 11 segments and 32 vertices, so this is precisely $P = \breve{F}(3^{11})$ in the convention of [1]. In particular, the number of segments is odd.

2 The orbit certificate and the contradiction

Theorem 1. *For the fence $P = \breve{F}(3^{11})$ defined above, the lower ideal $I_0 = \{5, 6, 7, 8, 12\}$ has a rowmotion orbit $\mathcal{O}$ satisfying*

$$|\mathcal{O}| = 237, \qquad \sum_{I \in \mathcal{O}} |I| = 3774.$$

Its least positive return time is 237. Consequently Conjecture 5.4(b) of [1] is false.

The complete finite certificate appears in Appendix A. An eight-digit hexadecimal integer m_k encodes

$$I_k = \{v \in \{1, \dots, 32\} : \text{bit } v-1 \text{ of } m_k \text{ is } 1\}.$$

The accompanying `data/orbit.csv` also lists the vertices, cardinality, and maximal antichain of every ideal. The following elementary certificate criterion explains exactly what is checked.

Lemma 1. *Let L be a positive integer. Suppose $I_0, \dots, I_{L-1}$ are distinct lower ideals and $\rho(I_k) = I_{(k+1) \bmod L}$ for every $0 \le k < L$. Then the forward orbit of I_0 is exactly this set of L ideals, and its least positive return time is L.*

Proof. Induction gives $\rho^n(I_0) = I_{n \bmod L}$ for every $n \ge 0$. Thus every iterate is in the list, and every listed ideal occurs as an iterate. Closure gives a return at L; distinctness excludes a return at any $0 < n < L$. □

Proof of Theorem 1. The principal lower sets of 5, 6, 7, 8, and 12 are contained in I_0, so I_0 is a lower ideal. For an explicit verification of the certificate, define $D(x) = \{y : y \le_P x\}$. The order can be evaluated without a closure algorithm: for $x < y$ numerically, $x \le_P y$ exactly when every edge between them points right, and $y \le_P x$ exactly when every such edge points left. Together with reflexivity this determines all principal lower sets.

For each of the 237 encoded sets, verify

$$x \in I_k \implies D(x) \subseteq I_k.$$

Then compute, with indices taken modulo 237,

$$M_k = \{x \notin I_k : D(x) \setminus \{x\} \subseteq I_k\}, \qquad I_{k+1} = \bigcup_{x \in M_k} D(x).$$

These formulas are respectively the lower-ideal condition, the minimal elements of the complement, and the definition of rowmotion. The supplied checker verifies all 237 equalities, including the last-to-first equality, distinctness of all 237 masks, and the cardinality sum 3774. The Lean development verifies the same finite conditions on actual finite sets and proves the conclusion of Lemma 1. All arithmetic is integral or rational.

It follows that the complete orbit has mean

$$\frac{3774}{237} = \frac{1258}{79} = 16 - \frac{6}{79} \neq 16 = \frac{|P|}{2}.$$

Equivalently, the conjectured sum is $237 \cdot 16 = 3792$, exceeding the actual sum by 18. This single orbit satisfies the conjecture's hypotheses and violates its conclusion. □

3 Reproduction and formal verification

From the root of the accompanying package, run

```
python checker/check.py
```

The checker uses only the Python standard library. It reads the supplied CSV instead of regenerating it, constructs the order directly from the path orientation, and checks the complete orbit and its agreement with the Lean orbit literals. The certificate is also fully printed here so that the finite data can be checked independently of the distribution.

The Lean development fixes Lean 4.19.0 and mathlib at commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

All transitive dependencies are pinned in the lockfile. Portable instructions, the actual build logs, and the axiom audit are included. The four modules have the following roles:

- `Poset`: the 32-element partial order; all maximal chains; distinct segments, shared vertices, and the parameter convention.
- `Rowmotion`: genuine lower ideals and the identity $\rho(I) = {\downarrow}\min(P \setminus I)$.
- `Orbit`: all 237 lower ideals, all transitions including closure, distinctness, and the sum 3774.
- `Main`: every iterate, completeness of the orbit, its least period and mean, and the failure of half-cardinality homomesy.

The theorem `W04.paper_fence_counterexample` combines the segment convention with the dynamical failure. The order on the vertices is the explicit fence relation, rather than the numerical order on `Fin 32`.

Trust assumptions. The finite orbit certificates use `native_decide`. The final counterexample theorem's printed axiom dependencies are

`propext, Classical.choice, Lean.ofReduceBool, Quot.sound`.

In particular, `Lean.ofReduceBool` adds trust in native evaluation and the correctness of the Lean compiler. This is not a completed kernel-reduction-only proof. The final sources contain no `sorry`, `admit`, or custom unproved mathematical axioms; the final theorem dependencies contain no `sorryAx`. The CSV and Python checker are not introduced as Lean axioms. Successful compilation is a machine verification result, not a claim of external human review.

4 Scope and literature boundary

The conclusion concerns Conjecture 5.4(b), about ideal-cardinality homomesy; it makes no assertion against Conjecture 5.4(a), about antichain-cardinality homometry. No minimality among counterexamples is claimed.

Mertin and Poznanović [2] restate the target as Conjecture 12 and study equivalent or stronger forms. Their three-segment result (Theorem 19) and their result for dihedral orbits (Theorem 26) do not assert the required average on each eleven-segment rowmotion orbit. The present counterexample does not depend on their identities. The literature check did not identify an earlier instance of this counterexample in the materials inspected; this is not an exhaustive priority determination. The official record and abstract of Mertin's 2025 dissertation [3] were accessible, but its full text could not be retrieved because the download returned HTTP 403. Its contents have not been ruled out as a prior source of the result. We therefore make no first-discovery claim.

AI assistance. This work was prepared with AI assistance. No independent human peer-review claim is made.

References

[1] S. Elizalde, M. Plante, T. Roby, and B. E. Sagan, *Rowmotion on fences*, Algebraic Combinatorics **6**(1) (2023), 17–36. doi:10.5802/alco.256.

[2] A. Mertin and S. Poznanović, *Toggleability Spaces of Fences*, Electronic Journal of Combinatorics **31**(4) (2024), P4.46. doi:10.37236/13085.

[3] A. Mertin, *Homomesies and Toggleability Spaces*, Ph.D. dissertation, Clemson University, August 2025. Official record 4090. Record and abstract inspected; full text not retrieved.

A The complete orbit in hexadecimal

Read the entries from left to right and then top to bottom, starting at $k = 0$. There are 237 entries, with eight entries per full line. Bit $v - 1$ encodes vertex v. Thus the first entry encodes $\{5, 6, 7, 8, 12\}$. The last entry maps back to the first under rowmotion. The sum of the numbers of set bits is 3774.

```
000008f0 20821c39 71c73e63 fbef80f0 3ffe0839 60031c63 f087bef0 39cffff9
63e00003 f0020820 38871c71 61cfbefb f3e3ffff 38060000 608f0821 f1c39c73
3be63ef8 7f8f0fe1 e0c39833 31e63c78 7b8f0ee1 ffc39ff3 00e63838 218f0c61
73c39ef3 f8e63ff8 018f0001 23c38823 70e61c70 f98f3ef9 03c38fe3 20e61830
718f3c79 fbc38ee3 3fe61ff0 600f3839 f0838c63 39c61ef0 63ef3ff9 f0ff8003
39800820 63c21c71 f0e73efb 398f8fff 63c01800 f0e23c21 39870e73 63cf98f8
f0e03c01 39820e23 63c71870 f0efbcf9 39fffe03 63800020 f0c20871 39e71cfb
638fbe3f f0c3f860 39e60cf1 638f1e3b f0c3b87f 39e7fce0 638e0e31 f0c3187b
39e7bcff 638ffe00 f0c00021 39e20873 63871cf8 f0cfbe01 39e3f823 63860c70
f0cf1ef9 39e3bfe3 6387f830 f0ce0c79 39e31ee3 6387bff0 f0cff839 39e00c63
63821ef0 f0c73ff9 39ef8003 63fe0820 f0031c71 3887befb 61cfffff f3e00000
38020821 60871c73 f1cfbef8 3be3ffe1 7f860033 e0cf0878 31e39ce1 7b863e33
ffcf0078 00e388e1 21861c33 73cf3e78 f8e380e1 01860833 23cf1c78 70e3bee1
f987fff3 03ce0038 20e30861 71879cf3 fbce3e38 3fe30061 600788f3 f08e1c38
39c33e61 63e780f3 f00e0838 38831c61 61c7bef3 f3efffff8 38fe0001 61830823
f3c79c70 38ee3ef9 61ff0fe3 f3839830 38c63c79 61ef0ee3 f3ff9ff0 38003839
60820c63 f1c71ef0 3befbff9 7ffff803 e0000c20 30821e71 79c738fb e3ef8c3f
30fe1e60 798338f1 e3c78c3b 30ee1e7f 79ff38e0 e3838c31 30c61e7b 79ef38ff
e3ff8c00 30001e21 78823873 e1c70cf8 33ef9e01 78fe3823 e1830c70 33c79ef9
78ee3fe3 e1ff0030 33838879 78c61ce3 e1ef3e30 33ff8079 780008e3 e0821c30
31c73e79 7bef80e3 fffe0830 00031c79 2087bee3 71cffff0 fbe00039 3f820863
60c71cf0 f1efbe39 3bfff863 7f800cf0 e0c21e39 31e73863 7b8f8cf0 ffc01e39
00e23863 21870cf0 73cf9e39 f8e03863 01820cf0 23c71e39 70efb863 f9fffcf0
03800e39 20c21863 71e73cf0 fb8f8e39 3fc01863 60e23cf0 f1870e39 3bcf9863
7fe03cf0 e0020e39 30871863 79cfbcf0 e3e3fe39 30060063 788f08f0 e1c39c39
33e63e63 780f00f0 e0838839 31c61c63 7bef3ef0 ffff8ff9 00001803 20823c20
71c70e71 fbef98fb 3ffe3c3f 60030e60 f08798f1 39ce3c3b 63e30e7f f00798e0
388e3c31 61c30e7b f3e798ff 380e3c00 60830e21 f1c79873 3bee3cf8 7fff0e01
e0039823 30863c70 79cf0ef9 e3e39fe3 30063830 788f0c79 e1c39ee3 33e63ff0
780f0039 e0838863 31c61cf0 7bef3e39 ffff8063
```